

Kush Ashvinbhai Patel

M.Tech – ICT (Machine Learning) | DA-IICT

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🌐 [LinkedIn](#)

🐙 [Github](#)

🌐 [Portfolio](#)

Education

M.Tech (ICT) in Machine Learning, Dhirubhai Ambani University (DA-IICT)

CPI: 9.39/10

2025 – Present

Gandhinagar, Gujarat

B.E (CE), Madhuben & Bhanubhai Patel Institute of Technology (CVMU)

CPI: 7.81/10

2021 – 2025

Anand, Gujarat

Experience

Information Retrieval & Language Processing (IRLP) Lab, DAU

Graduate Research Assistant – Supervised by Prof. T. Bandyopadhyay (DG, DAU)

Sep 2025 – Present

Gandhinagar, Gujarat

- Researching **post-training fine-tuning**, **safety guardrails**, and **dense/sparse retrieval** on 100K+ text corpora.
- Benchmarking **transformer models** for code-mixed classification and **Recall@K / NDCG** with **90%+ PyTest**.

HP Param IT Solutions

Software Engineer Intern

Jun 2024 – Jul 2025

Remote

- Engineered scalable ETL data pipelines and REST APIs on **AWS (EC2, S3)**, handling **50K+ daily queries**.
- Designed **PostgreSQL** schemas, composite indexes, and analytical SQL queries, cutting latency by **35%**.

Projects

Beyond Visible Spectrum: Foundation Model Adaptation | PyTorch, Swin Transformer, timm [[paper](#) | [github](#) | [kaggle](#)]

- Published at ICPR 2026 (Springer LNCS, pp. 276–290); ranked **3rd globally** out of 574 participants.
- Adapted Swin Transformer with 5-fold CV softmax ensembling, achieving **89.3% accuracy (0.79 Macro F1)**.

GSSTB Scholar: Agentic System & Hallucination Mitigation Engine | FastAPI, LangGraph, ChromaDB, Docker [[live](#) | [github](#)]

- Architected an enterprise **6-node LangGraph** agentic platform with **hybrid semantic search** (dense + BM25).
- Engineered automated **hallucination mitigation gates**, output safety guardrails, and **Docker** CI/CD.

ShelfMind AI: Retail Intelligence & Vector Embedding Engine | YOLO26s, DINOv2, FAISS, Streamlit [[github](#) | [demo](#)]

- Achieved **58.3 mAP@50-95** on **SKU-110K**, outperforming published baselines with **7x fewer parameters**.
- Engineered a **DINOv2 + FAISS** vector similarity pipeline (256-d embeddings, <50ms latency) on **HF Spaces**.

Technical Skills

Foundation Models & Alignment: PyTorch, HuggingFace, Transformers (BERT, LLaMA, Swin), Fine-Tuning, Post-Training

Agentic AI & Safety: LangGraph, Agentic Workflows, Tool Calling, Hallucination Mitigation, Guardrails, Hybrid RAG

Tabular ML & Optimization: LightGBM, XGBoost, CatBoost, Scikit-learn, FAISS, DINOv2, Uncertainty Quantification

Systems, Cloud & Tooling: Python, SQL, C/C++, Docker, AWS (EC2, S3), FastAPI, Git/GitHub, Linux Shell, PyTest

Achievements

- **ICPR 2026 (Published):** Published in Springer LNCS (pp. 276–290); ranked **3rd globally** (574 participants). [[paper](#)]
- **Kaggle – Bidding Predictions:** Ranked **1st Public / 2nd Private** using LightGBM, XGBoost, CatBoost on 832K rows.
- **Amazon ML Summer School 2026:** Selected among **top 3,000** across India for Amazon's flagship ML programme.
- **GATE Qualified (Dual):** Cleared GATE in **Computer Science (Score: 456)** and **Data Science & AI (Score: 413)**.
- **Competitive Programming:** [LeetCode](#) – 1682 (Top 22%); [CodeChef](#) – 3-Star, 1661; [Codeforces](#) – 1451 (Specialist).
- **Teaching Assistant (DSA) & Leadership:** Mentored **30+ students** in C++ DSA; official **Google Gemini Student Ambassador**.
- **Certifications:** AI Agents Intensive (Google & Kaggle), Machine Learning (IBM), SQL (Michigan), PyTorch (DL.AI).